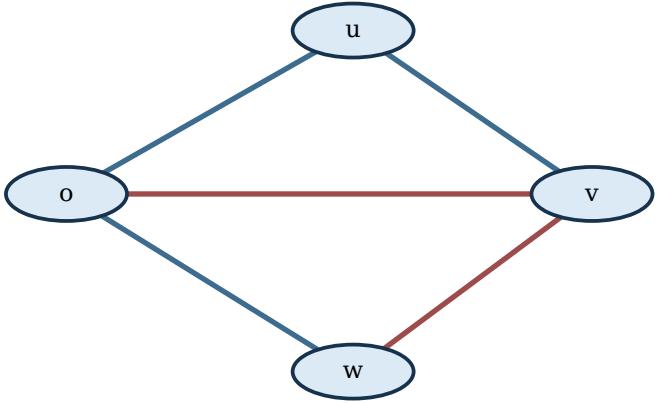
Finite holonomy selects the greatest path-independent subfamily

Tree gauge + non-tree cycle edges



blue: spanning tree red: fundamental cycles

Fundamental holonomies
$H_e = G_v^{-1} T_e G_u$

Stacked defect map
$\Delta_S x = ((H_e - I)x)_e$

Maximal path-flat core
$C_o = \ker \Delta_S$

$\dim C_o = \dim F_o - \operatorname{rank} \Delta_S$

Full global frame iff every $H_e = I$